Thesis Update #1

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Project Motivation

- Diabetes requires frequent glucose monitoring
- Finger-prick testing is accurate but inconvenient
- Goal: build a painless wearable system for real-time glucose estimation

Project Objectives

- Review prior work on non-invasive glucose monitoring
- Research NIR-spectroscopy
- Identify hardware/software for first prototype
- Define early validation Plan

Research Review

- Non-invasive Glucose Detection Methods
  - A deep learning model for inference on PPG signals was tested
  - Achieved RSME = 19.7 mg/dL and 100% Clarke Error Grid A+B zones
  - Improve accuracy and robustness of blood glucose estimation
  - Current work moves toward hardware development

NIR-spectroscopy

- Absorbance-based spectroscopy
  - Near-Infrared waves penetrate material
  - Molecular bonds absorb IR at different wavelengths
  - Wavelength absorbed depends on atomic weight and bond strength
  - Detector measures absorbed NIR spectrum [1]

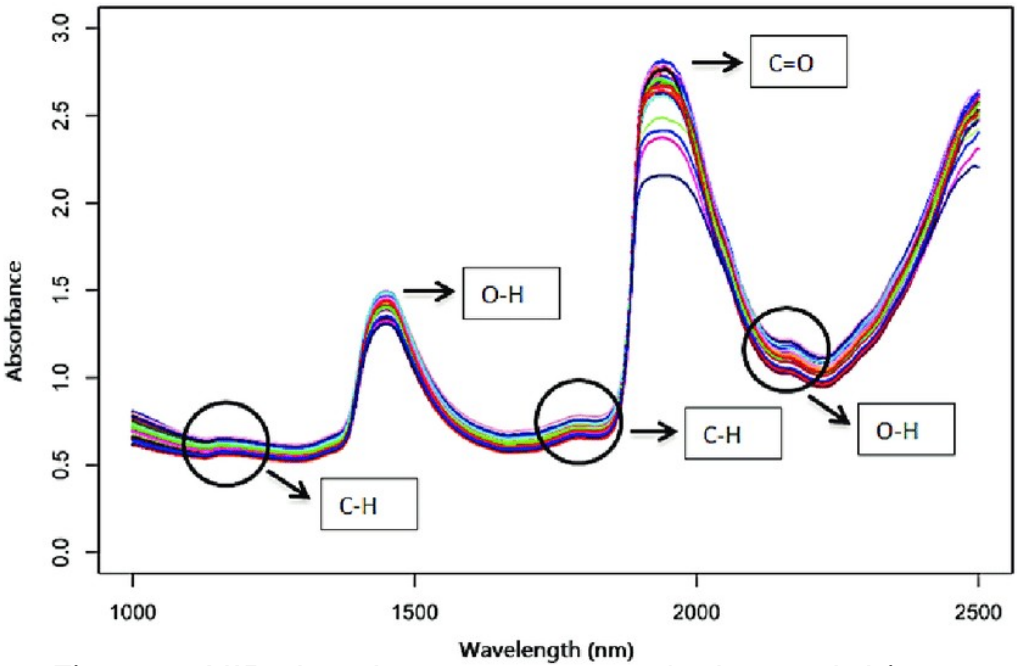


Figure 1: NIR absorbance spectrum for hemoglobin, adapted from [1]

Transmission vs Reflectance Mode

- Transmission
  - Light forward scatters through tissue; detector is opposite the source
  - Used for determining the best wavelength for glucose detection
- Reflectance
  - Light back scatters from tissue; detector is beside the source
  - Used for the wearable prototype

NIR for Glucose

- Glucose contains O-H, C-H, and C-O bonds
- Absorption changes for these bonds can be detected
- Detection difficult due to the presence of water and hemoglobin
- Need to determine the wavelength that provides the most accurate glucose detection

First Prototype

- Analog Discovery 3 ($374)
- ESP32-S3 ($15)
  - 2 MB RAM
  - Wi-Fi and Bluetooth
  - Dual-Core Xtensa vector processor
  - Compare vector vs normal processor

References

[1] U. Nasiba, R. P. Jenie, Irzaman, and H. Alatas, “Determination of wavelength candidates for non-invasive hemoglobin measurement devices and energy spectrum analysis,” *AIP Conference Proceedings*, vol. 2194, no. 1, Art. no. 020069, Dec. 2019, doi: 10.1063/1.5139801.